Which retrieval settings find the true value?

Truth coverage and graded relevance of the documents medsim retrieves (Europe PMC + LitSense), on the 141 set A questions of the retrieval benchmark · judge anthropic/claude-haiku-5.5 · 37 scored configurations and controls

> **Result.** The best configuration is **llm_select20+epmc25+q_synonym** (an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50); Europe PMC MeSH synonym expansion): strict truth coverage 70.2% against the baseline's 60.3% (+9.9 (+3.5 to +17.0) points, paired over the same questions), nDCG@8 0.750 against 0.691, and precision@8 49.6% against 41.3%. On fresh queries: query set fresh1: strict +7.1 points, nDCG +0.054; query set fresh2: strict +12.1 points, nDCG +0.058.

What was measured

- **Strict truth coverage** (decides): a document gives a value or range for this variable in patients with this or a closely related condition, and the hidden true value lies inside it or within about 15%. **Category coverage** (shown): a document's value, range, or level, a reference range for comparable patients included, puts the true value in the same low / normal / high category.
- **Graded relevance**: each document 0 (not about the variable or the condition), 1 (about them, no usable value), or 2 (a value for the variable in comparable patients). **nDCG@8** is against a perfect list of eight grade-2 documents, so it rewards better documents and a better order (it breaks ties); **precision@8** is the share of grade-2 documents.
- One judge call per question grades all eight documents and gives both coverage verdicts. Every positive verdict needs a verbatim quote that code finds in the named document; an unverified grade 2 counts as 1, and unverified coverage as not covered.
- Each configuration retrieves for the same **frozen search query** per question (medsim's Stages A and B once, with the stored runs' model deepseek/deepseek-v4-flash-0731), and no answer is written. Differences between configurations therefore come from the retrieval settings. Eight documents per question throughout. Search results are cached, so a configuration that issues the same requests gets the same results.

Can the metrics be trusted?

Check	Strict coverage	Category	nDCG@8	Reading
Shuffled control: each question judged on another question's documents	0.0%	0.7%	0.034	Near zero: the judge reads the documents rather than answering from the question.
The baseline's documents judged twice	60.3% / 61.7%	83.7% / 85.8%	0.691 / 0.680	The judge's own noise: 1.4 points of strict coverage.
Stored benchmark runs 2–6 of the baseline method (fresh queries, full pipeline)	61.7% ± 3.4	–	0.675 ± 0.019	Run-to-run spread (mean ± SD) when the query is written anew; the frozen-query baseline (60.3%) lies inside it.
Stored run 2 of the original retrieval (`current`)	40.4%	70.2%	0.593	About 20 points below changes 1–4: the metrics separate a known difference.

Single changes from the baseline

The baseline is medsim's default retrieval (changes 1–4: value-first ranking in one list, 50 Europe PMC candidates, animal and age filter, article-body search with excerpts, value-based broadening). Each row changes one setting. Differences are paired over the same 141 questions, with 95% bootstrap intervals; green intervals exclude zero.

Configuration	Strict coverage	Δ strict vs baseline (95% CI)	Category coverage	nDCG@8	Δ nDCG vs baseline	Precision@8	n
baseline baseline: changes 1-4	60.3%		83.7%	0.691		41.3%	141
llm_select40 an LLM picks the 8 documents from the best 40	68.8%	+8.5 (+2.1 to +15.6)	89.4%	0.746	+0.055 (+0.030 to +0.078)	50.8%	141
llm_select20 an LLM picks the 8 documents from the best 20	67.4%	+7.1 (+0.7 to +14.2)	89.4%	0.749	+0.058 (+0.041 to +0.076)	49.6%	141
epmc25 25 Europe PMC candidates (baseline 50)	66.0%	+5.7 (-1.4 to +12.8)	85.1%	0.690	-0.001 (-0.020 to +0.017)	42.6%	141
q_synonym Europe PMC MeSH synonym expansion	63.8%	+3.5 (-1.4 to +8.5)	84.4%	0.690	-0.001 (-0.015 to +0.013)	42.3%	141
epmc_only Europe PMC only	63.1%	+2.8 (-2.8 to +8.5)	85.8%	0.665	-0.026 (-0.048 to -0.005)	40.5%	141
ls_sentences LitSense sentences instead of passages	63.1%	+2.8 (-4.3 to +9.9)	86.5%	0.659	-0.032 (-0.053 to -0.013)	39.3%	141
q_natural LitSense natural-language queries	62.4%	+2.1 (-4.3 to +8.5)	83.7%	0.684	-0.007 (-0.029 to +0.014)	41.3%	141
round_robin round-robin merge across sources instead of one list	62.4%	+2.1 (-4.3 to +8.5)	83.7%	0.667	-0.024 (-0.044 to -0.006)	39.6%	141
ls100 100 LitSense results	61.7%	+1.4 (-5.0 to +7.8)	86.5%	0.681	-0.010 (-0.028 to +0.007)	40.9%	141
epmc100 100 Europe PMC candidates	61.7%	+1.4 (-5.0 to +7.8)	85.8%	0.679	-0.012 (-0.028 to +0.004)	40.5%	141
fulltext20 20 open-access full texts fetched for excerpts (baseline 10)	61.0%	+0.7 (-5.7 to +7.1)	87.9%	0.700	+0.009 (-0.008 to +0.026)	44.7%	141
ls60 60 LitSense results (baseline 30)	61.0%	+0.7 (-5.7 to +7.1)	86.5%	0.681	-0.010 (-0.025 to +0.005)	41.2%	141
ls_no_rerank no LitSense-side rerank	61.0%	+0.7 (-5.7 to +7.1)	87.2%	0.677	-0.014 (-0.033 to +0.004)	40.5%	141
chars3000 3,000 characters per document (baseline 1,500)	60.3%	+0.0 (-6.4 to +6.4)	86.5%	0.677	-0.013 (-0.035 to +0.006)	41.7%	141
no_value_first no value-first ranking	59.6%	-0.7 (-7.1 to +5.7)	84.4%	0.683	-0.008 (-0.026 to +0.009)	41.5%	141
ladder_v1 the original broadening ladder (v1)	59.6%	-0.7 (-7.8 to +6.4)	85.8%	0.661	-0.029 (-0.050 to -0.010)	37.8%	141
q_gpt6luna openai/gpt-6-luna writes the search query	58.9%	-1.4 (-10.6 to +7.1)	87.9%	0.704	+0.014 (-0.016 to +0.040)	44.0%	141
no_population_filter no animal and age-group filter	56.7%	-3.5 (-9.2 to +2.1)	85.8%	0.682	-0.008 (-0.026 to +0.008)	40.7%	141
no_excerpts no full-text excerpts (abstracts and passages only)	39.0%	-21.3 (-29.8 to -12.8)	70.9%	0.597	-0.094 (-0.122 to -0.068)	29.7%	141
no_lexical_rerank no lexical rerank within each source	34.8%	-25.5 (-34.0 to -17.0)	66.0%	0.554	-0.136 (-0.164 to -0.110)	20.3%	141
litsense_only LitSense only	29.8%	-30.5 (-40.4 to -21.3)	61.7%	0.576	-0.115 (-0.144 to -0.087)	23.8%	141

With 21 changes tested at 95%, about one could look better or worse by chance alone; treat single changes whose interval only just excludes zero with care.

Combinations

The changes with the largest positive estimates, added to the best single change.

Configuration	Strict coverage	Δ strict vs baseline (95% CI)	Category coverage	nDCG@8	Δ nDCG vs baseline	Precision@8	n
baseline baseline: changes 1-4	60.3%		83.7%	0.691		41.3%	141
llm_select20+epmc25+q_synonym **an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50); Europe PMC MeSH synonym expansion**	**70.2%**	**+9.9** (+3.5 to +17.0)	**87.9%**	**0.750**	**+0.059** (+0.042 to +0.076)	**49.6%**	**141**
llm_select20+q_synonym an LLM picks the 8 documents from the best 20; Europe PMC MeSH synonym expansion	68.8%	+8.5 (+2.8 to +14.9)	90.8%	0.743	+0.052 (+0.035 to +0.070)	48.8%	141
llm_select20+epmc25 an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50)	68.8%	+8.5 (+1.4 to +14.9)	87.9%	0.737	+0.046 (+0.027 to +0.065)	48.3%	141

Confirmation on fresh queries

The baseline and the best configuration again, each on two new sets of queries written by the same query builder: the gain should hold when the queries change. Here each Δ compares with the baseline on the same query set.

Configuration	Strict coverage	Δ strict vs baseline (95% CI)	Category coverage	nDCG@8	Δ nDCG vs baseline	Precision@8	n
baseline@fresh1 baseline: changes 1-4 (queries: fresh1)	62.4%		87.9%	0.681		40.7%	141
llm_select20+epmc25+q_synonym@fresh1 **an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50); Europe PMC MeSH synonym expansion (queries: fresh1)**	**69.5%**	**+7.1** (-0.7 to +14.9)	**89.4%**	**0.735**	**+0.054** (+0.036 to +0.074)	**47.1%**	**141**
baseline@fresh2 baseline: changes 1-4 (queries: fresh2)	57.4%		85.1%	0.682		41.7%	141
llm_select20+epmc25+q_synonym@fresh2 **an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50); Europe PMC MeSH synonym expansion (queries: fresh2)**	**69.5%**	**+12.1** (+4.3 to +19.9)	**92.9%**	**0.740**	**+0.058** (+0.039 to +0.078)	**47.5%**	**141**

Faithfulness of the answers written on these documents

medsim's answer-writing stage (Stage C) was run on each configuration's documents, with the frozen queries, and every answer with documents was judged for faithfulness with the benchmark's rubric (`bench.answer_metrics`): does at least one document give a value for this variable in a comparable patient that agrees with the answer? Same judge; a question with no answer counts as not supported.

Configuration	Supported (95% CI)	Δ vs baseline (paired)	Covered and supported	Covered, not supported	Not covered, supported	Neither
baseline	22.0% (15.6 to 29.1)		23	62	8	48
llm_select20+epmc25+q_synonym	**32.6%** (25.5 to 40.4)	**+10.6** (+2.8 to +19.1)	**35**	**64**	**11**	**31**

"Covered" is strict truth coverage of the same documents. Covered but not supported means the documents held a value matching the truth but the answer was not backed by them: the answer-writing stage, not retrieval, is where those questions are lost. This judge reads "agrees" strictly (in one example, 9.8 cm in the document against 9.6 cm in the answer was not supported).

Cost and limits

- Cost of this work from the stored usage records: $9.69 (judge $7.15, LLM selection during retrieval $1.79, query writing $0.32). Europe PMC and LitSense are free.
- One judge, an LLM, without human labels: absolute levels may be off; the comparisons between configurations, made by the same judge on the same questions, are the point.
- 141 questions: differences of a few points are inside the noise.

- Europe PMC answered HTTP 503 during the first retrieval of the two fresh-query baseline runs (129 and 64 of 141 questions lost it). Those questions were retrieved again once it recovered and judged again; the tables use the second attempt. Every other run had no failed source.
- LLM selection adds one call per question to medsim's retrieval (about $0.0016 with deepseek/deepseek-v4-flash-0731, and its latency). It is not medsim's default yet.
- The best configuration was chosen on these questions; the fresh-query runs re-test it, but on the same 141 questions, not on new ones.